

Faraday Rotation

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I. Introduction

More than 150 years ago, Michael Faraday discovered that *linearly polarized* light traveling through a substance experiences a rotation when a magnetic field is applied to the material. The amount of rotation (φ) was found to be proportional to the magnitude of the magnetic field ($\mathbf{B}$) and to the length of the sample (L),

$$\varphi = V B L,$$

where the constant of proportionality (V) is called the Verdet constant.

A linearly polarized beam of light, one that has a unique electric field E -vector direction, can be decomposed into two circularly polarized beams of equal intensity. These right- and left-circularly polarized beams propagate simultaneously, but are independent of each other (orthogonal). An applied B -field causes the material to become *optically active* – producing different refractive indices for the two beams so that they travel at different phase velocities. Thus, when the two beams exit the material they have a different phase relationship, which is manifest in a rotation of the E -vector of the combined beams.

In this experiment you will set up an apparatus for measuring very small rotations ($< 10^{-3}$ deg) of polarized light passing through samples placed in a magnet solenoid. The Verdet constant will be determined for glass and water.

II. Apparatus (be sure to write down model numbers in your report)

- optical breadboard, 2 Glan (calcite) polarizers with rotating mounts
- 2 silicon photodiodes, collimated laser diode, GW frequency generator
- SRS lock-in amplifier, 3 DC voltmeters
- 20 mm cuvette cells with 1.1 mm thick BK-7 glass ($\sim\text{SiO}_2$) windows

III. Procedure

A. Calibrate Magnetic Field

Connect power supply to electromagnet solenoid *through* a current meter – don't exceed **3.7 amps** (15V). With the Hall-probe gaussmeter, measure the B -field in the magnet center at a few current values (I).

Plot $B(I)$ and fit to get the coefficient α , from $B = \alpha I$.

B. Initial Setup

Make sure the GW frequency generator is **OFF** before connecting/disconnecting the laser diode.

Connect the TTL/CMOS output of the GW frequency generator to the laser diode.

Connect the generator to the lock-in reference (REF).

Pull out the CMOS knob and rotate to minimum (full CCW).

Turn **ON** generator, set to $f=500$ Hz, and

increase amplitude to maximum (CW) while looking for red spot on a card.

You are now ready to align the optical components.

C. Alignment of Optics

Remove both polarizers.

Using a white card, position the laser so that the beam is horizontally parallel to the breadboard plate, and passes through the magnet a few mm below the centerline.

Adjust the photodiode PD-1 so that the beam hits its center.

Place polarizer POL-2 in the beam and align so that the laser beam passes through its center.

Rotate POL-2 so that the *side exit beam is parallel to breadboard* and hits the diode PD-2.

Place polarizer POL-1 in the beam and align so that the laser passes through the center of both polarizers.

Check that the beam exiting POL-1 is polarized at approximately 45 deg from vertical.

Adjust the position of diodes PD-1 and PD-2 so that the laser beam is centered on them.

Rotate the angle on POL-1 to get equal voltages from the PDs using DC voltmeters.

What are the polarization states of the two beams exiting POL-2?

What does the output of PD-1 look like on the scope? Compare to voltmeter value and explain.

The polarization is defined as the voltage difference of the 2 PDs divided by their sum,

$$\rho \equiv V_- / V_+ = (V_1 - V_2) / (V_1 + V_2).$$

D. Configure Lock-in Amplifier

Connect the PD outputs to the differential inputs of the lock-in amplifier, and to 2 DC voltmeters.

Use the output of the frequency generator as the lock-in reference.

Configure the lock-in to display **R** and **Theta** for V_- (set A-B). See Appendix for lock-in information.

E. Calibrate Rotation (see Appendix)

Make a table containing the columns: D , R , θ , V_1 , V_2 , $V_1 - V_2$, $V_1 + V_2$, and $\rho = R / (V_1 + V_2)$.

Rotate POL-1 in the mount until the magnitude **R** is near zero.

Rotate POL-1 in small steps ($\Delta D = 0.001''$) to 3-4 positions on both sides and compute $\rho(D)$.

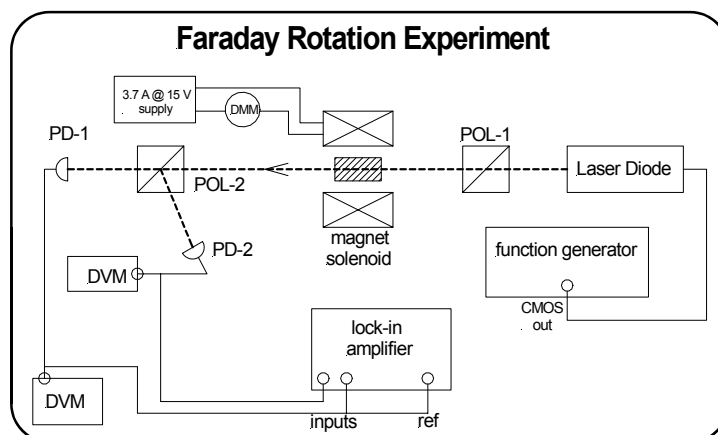
Note that R is a magnitude, so when $V_1 - V_2$ changes sign, the angle θ will change by nearly 180 deg.

Question: Is R is about equal to $V_1 - V_2$?

The FR angle is given by $\varphi = \beta \rho$, where β is a constant for the setup.

Plot and curve fit $\rho(D)$. Compute $\beta = d\varphi/d\rho$ from the slope $d\rho/dD$ and $d\varphi/dD$ (see appendix).

What is the noise (error) level of the lock-in in equivalent μrad ?



F. Measure Faraday Rotation

Make a table in your notebook with columns: I , V_1 , V_2 , R , θ , V_+ , and $\rho = R / (V_1 + V_2)$.

Water + Glass Cell

Record the values V_1 and V_2 from the voltmeters, and R and θ from the lock-in.

Place the water-filled cell in the center of the magnet.

Re-record the values V_1 and V_2 from the voltmeters, and R and θ from the lock-in.

Did R or V_+ change? Explain.

Record ρ_{WG} while varying I for several positive and negative I -values up to $|I| = 3.7$ A.

Plot $\rho_{WG}(I)$. Curvefit $\rho_{WG}(I)$ to obtain $d\rho_{WG}/dI$.

Glass Cell Only

Set the magnet current to $I=0$.

Remove the water-filled cell and replace with the empty cell.

Re-record the values V_1 and V_2 from the voltmeters, and R and θ from the lock-in.

Did R or V_+ change? Explain.

Record ρ_G while varying I for several positive and negative I -values up to $|I| = 3.7$ A.

Plot $\rho_G(I)$. Curvefit $\rho_G(I)$ to obtain $d\rho_G/dI$.

Verdet Constants

Compute the Verdet constant and uncertainty for BK-7 glass from $V_G = \frac{\beta}{\alpha} \frac{1}{L_G} \frac{d\rho_G}{dI}$.

Compute the Verdet constant and uncertainty for water only from $V_W = \frac{\beta}{\alpha} \frac{1}{L_W} \left[\frac{d\rho_{WG}}{dI} - \frac{d\rho_G}{dI} \right]$.

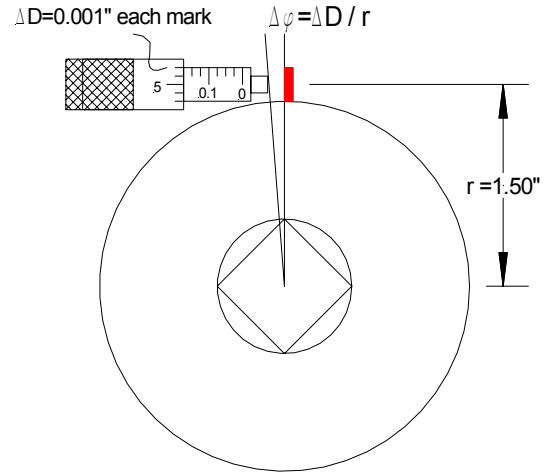
Compare your values to expected values.

Put these values in a table in your report and discuss.

IV. Appendix: Calibrating Faraday Rotation Angle

Here are notes for calibrating the Faraday rotation setup. They are used to determine the coefficient $\beta = d\phi/d\rho$, which relates the FR angle ϕ to the measured polarization ρ .

First, note that the angle of the linear polarizer POL-1 is rotated by the micrometer. The angle of the polarizer ϕ is a function of the micrometer setting D . The calcite polarizer is installed in a rotating mount which has two modes of rotation, gross and fine. Do the gross adjustment only if the polarization is not approximately set to pass polarization at 45 deg from the vertical. Gross rotation is achieved by unscrewing the small lower locking screw about 1/4 turn, then rotating the whole barrel. Fine rotation is accomplished by tightening the locking screw then turning the micrometer on the top. Notice that each mark on the turning barrel of the micrometer corresponds to $\Delta D = 0.001''$ movement of the micrometer plunger. Also notice that the micrometer plunger is $r = 1.5''$ from the center of the polarizer. D and r are related by the formula $\Delta D = r \Delta\phi$, where D and r are in the same units and ϕ has units of radians (57.3 deg/rad).



For calibration, the polarization $\rho(D)$ is found by determining ρ for several values of D , differing by $\Delta D = 0.001''$, on either side of $\rho = 0$. Then, $d\rho/dD$ is determined from the slope of the $\rho(D)$ plot. Rotating the micrometer by one mark ($\Delta D = 0.001''$) corresponds to a rotation of $\phi = 667$ microradians, so that

$$\frac{d\phi}{dD} = 667 \frac{\mu rad}{0.001''}.$$

Finally, the coefficient β is computed from the measured $d\rho/dD$ using

$$\beta = \frac{d\phi}{dD} \times \frac{1}{d\rho/dD}.$$

WHAT IS A LOCK-IN AMPLIFIER?

Lock-in amplifiers are used to detect and measure very small AC signals - all the way down to a few nanovolts! Accurate measurements may be made even when the small signal is obscured by noise sources many thousands of times larger.

Lock-in amplifiers use a technique known as phase-sensitive detection to single out the component of the signal at a specific reference frequency AND phase. Noise signals at frequencies other than the reference frequency are rejected and do not affect the measurement.

Why use a lock-in?

Let's consider an example. Suppose the signal is a 10 nV sine wave at 10 kHz. Clearly some amplification is required. A good low noise amplifier will have about 5 nV/ $\sqrt{\text{Hz}}$ of input noise. If the amplifier bandwidth is 100 kHz and the gain is 1000, then we can expect our output to be 10 μV of signal (10 nV x 1000) and 1.6 mV of broadband noise (5 nV/ $\sqrt{\text{Hz}}$ x $\sqrt{100 \text{ kHz}}$ x 1000). We won't have much luck measuring the output signal unless we single out the frequency of interest.

If we follow the amplifier with a band pass filter with a $Q=100$ (a VERY good filter) centered at 10 kHz, any signal in a 100 Hz bandwidth will be detected (10 kHz/ Q). The noise in the filter pass band will be 50 μV (5 nV/ $\sqrt{\text{Hz}}$ x $\sqrt{100 \text{ Hz}}$ x 1000) and the signal will still be 10 μV . The output noise is much greater than the signal and an accurate measurement can not be made. Further gain will not help the signal to noise problem.

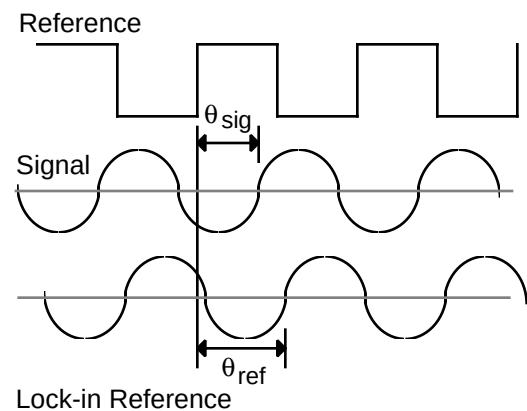
Now try following the amplifier with a phase-sensitive detector (PSD). The PSD can detect the signal at 10 kHz with a bandwidth as narrow as 0.01 Hz! In this case, the noise in the detection bandwidth will be only 0.5 μV (5 nV/ $\sqrt{\text{Hz}}$ x $\sqrt{0.01 \text{ Hz}}$ x 1000) while the signal is still 10 μV . The signal to noise ratio is now 20 and an accurate measurement of the signal is possible.

What is phase-sensitive detection?

Lock-in measurements require a frequency reference. Typically an experiment is excited at a fixed frequency (from an oscillator or function generator) and the lock-in detects the response from the

experiment at the reference frequency. In the diagram below, the reference signal is a square wave at frequency ω_r . This might be the sync output from a function generator. If the sine output from the function generator is used to excite the experiment, the response might be the signal waveform shown below. The signal is $V_{\text{sig}} \sin(\omega_r t + \theta_{\text{sig}})$ where V_{sig} is the signal amplitude.

The SR830 generates its own sine wave, shown as the lock-in reference below. The lock-in reference is $V_L \sin(\omega_L t + \theta_{\text{ref}})$.



The SR830 amplifies the signal and then multiplies it by the lock-in reference using a phase-sensitive detector or multiplier. The output of the PSD is simply the product of two sine waves.

$$V_{\text{psd}} = V_{\text{sig}} V_L \sin(\omega_r t + \theta_{\text{sig}}) \sin(\omega_L t + \theta_{\text{ref}})$$

$$= \frac{1}{2} V_{\text{sig}} V_L \cos([\omega_r - \omega_L]t + \theta_{\text{sig}} - \theta_{\text{ref}}) - \frac{1}{2} V_{\text{sig}} V_L \cos([\omega_r + \omega_L]t + \theta_{\text{sig}} + \theta_{\text{ref}})$$

The PSD output is two AC signals, one at the difference frequency ($\omega_r - \omega_L$) and the other at the sum frequency ($\omega_r + \omega_L$).

If the PSD output is passed through a low pass filter, the AC signals are removed. What will be left? In the general case, nothing. However, if ω_r equals ω_L , the difference frequency component will be a DC signal. In this case, the filtered PSD output will be

$$V_{\text{psd}} = \frac{1}{2} V_{\text{sig}} V_L \cos(\theta_{\text{sig}} - \theta_{\text{ref}})$$

This is a very nice signal - it is a DC signal proportional to the signal amplitude.

Narrow band detection

Now suppose the input is made up of signal plus noise. The PSD and low pass filter only detect signals whose frequencies are very close to the lock-in reference frequency. Noise signals at frequencies far from the reference are attenuated at the PSD output by the low pass filter (neither $\omega_{\text{noise}} - \omega_{\text{ref}}$ nor $\omega_{\text{noise}} + \omega_{\text{ref}}$ are close to DC). Noise at frequencies very close to the reference frequency will result in very low frequency AC outputs from the PSD ($|\omega_{\text{noise}} - \omega_{\text{ref}}|$ is small). Their attenuation depends upon the low pass filter bandwidth and roll-off. A narrower bandwidth will remove noise sources very close to the reference frequency, a wider bandwidth allows these signals to pass. The low pass filter bandwidth determines the bandwidth of detection. Only the signal at the reference frequency will result in a true DC output and be unaffected by the low pass filter. This is the signal we want to measure.

Where does the lock-in reference come from?

We need to make the lock-in reference the same as the signal frequency, i.e. $\omega_r = \omega_L$. Not only do the frequencies have to be the same, the phase between the signals can not change with time, otherwise $\cos(\theta_{\text{sig}} - \theta_{\text{ref}})$ will change and V_{psd} will not be a DC signal. In other words, the lock-in reference needs to be phase-locked to the signal reference.

Lock-in amplifiers use a phase-locked-loop (PLL) to generate the reference signal. An external reference signal (in this case, the reference square wave) is provided to the lock-in. The PLL in the lock-in locks the internal reference oscillator to this external reference, resulting in a reference sine wave at ω_r with a fixed phase shift of θ_{ref} . Since the PLL actively tracks the external reference, changes in the external reference frequency do not affect the measurement.

All lock-in measurements require a reference signal.

In this case, the reference is provided by the excitation source (the function generator). This is called an external reference source. In many situations, the SR830's internal oscillator may be used instead. The internal oscillator is just like a function generator (with variable sine output and a TTL

sync) which is always phase-locked to the reference oscillator.

Magnitude and phase

Remember that the PSD output is proportional to $V_{\text{sig}} \cos \theta$ where $\theta = (\theta_{\text{sig}} - \theta_{\text{ref}})$. θ is the phase difference between the signal and the lock-in reference oscillator. By adjusting θ_{ref} we can make θ equal to zero, in which case we can measure V_{sig} ($\cos \theta = 1$). Conversely, if θ is 90° , there will be no output at all. A lock-in with a single PSD is called a single-phase lock-in and its output is $V_{\text{sig}} \cos \theta$.

This phase dependency can be eliminated by adding a second PSD. If the second PSD multiplies the signal with the reference oscillator shifted by 90° , i.e. $V_L \sin(\omega_L t + \theta_{\text{ref}} + 90^\circ)$, its low pass filtered output will be

$$V_{\text{psd2}} = 1/2 V_{\text{sig}} V_L \sin(\theta_{\text{sig}} - \theta_{\text{ref}})$$

$$V_{\text{psd2}} \sim V_{\text{sig}} \sin \theta$$

Now we have two outputs, one proportional to $\cos \theta$ and the other proportional to $\sin \theta$. If we call the first output X and the second Y,

$$X = V_{\text{sig}} \cos \theta \quad Y = V_{\text{sig}} \sin \theta$$

these two quantities represent the signal as a vector relative to the lock-in reference oscillator. X is called the 'in-phase' component and Y the 'quadrature' component. This is because when $\theta=0$, X measures the signal while Y is zero.

By computing the magnitude (R) of the signal vector, the phase dependency is removed.

$$R = (X^2 + Y^2)^{1/2} = V_{\text{sig}}$$

R measures the signal amplitude and does not depend upon the phase between the signal and lock-in reference.

A dual-phase lock-in, such as the SR830, has two PSD's, with reference oscillators 90° apart, and can measure X, Y and R directly. In addition, the phase θ between the signal and lock-in reference, can be measured according to

$$\theta = \tan^{-1} (Y/X)$$

WHAT DOES A LOCK-IN MEASURE?

So what exactly does the SR830 measure? Fourier's theorem basically states that any input signal can be represented as the sum of many, many sine waves of differing amplitudes, frequencies and phases. This is generally considered as representing the signal in the "frequency domain". Normal oscilloscopes display the signal in the "time domain". Except in the case of clean sine waves, the time domain representation does not convey very much information about the various frequencies which make up the signal.

What does the SR830 measure?

The SR830 multiplies the signal by a pure sine wave at the reference frequency. All components of the input signal are multiplied by the reference simultaneously. Mathematically speaking, sine waves of differing frequencies are orthogonal, i.e. the average of the product of two sine waves is zero unless the frequencies are EXACTLY the same. In the SR830, the product of this multiplication yields a DC output signal proportional to the component of the signal whose frequency is exactly locked to the reference frequency. The low pass filter which follows the multiplier provides the averaging which removes the products of the reference with components at all other frequencies.

The SR830, because it multiplies the signal with a pure sine wave, measures the single Fourier (sine) component of the signal at the reference frequency. Let's take a look at an example. Suppose the input signal is a simple square wave at frequency f . The square wave is actually composed of many sine waves at multiples of f with carefully related amplitudes and phases. A 2V pk-pk square wave can be expressed as

$$S(t) = 1.273\sin(\omega t) + 0.4244\sin(3\omega t) + 0.2546\sin(5\omega t) + \dots$$

where $\omega = 2\pi f$. The SR830, locked to f will single out the first component. The measured signal will be $1.273\sin(\omega t)$, not the 2V pk-pk that you'd measure on a scope.

In the general case, the input consists of signal plus noise. Noise is represented as varying signals at all frequencies. The ideal lock-in only responds to noise at the reference frequency. Noise at other

frequencies is removed by the low pass filter following the multiplier. This "bandwidth narrowing" is the primary advantage that a lock-in amplifier provides. Only inputs at frequencies at the reference frequency result in an output.

RMS or Peak?

Lock-in amplifiers as a general rule display the input signal in Volts RMS. When the SR830 displays a magnitude of 1V (rms), the component of the input signal at the reference frequency is a sine wave with an amplitude of 1 Vrms or 2.8 V pk-pk.

Thus, in the previous example with a 2 V pk-pk square wave input, the SR830 would detect the first sine component, $1.273\sin(\omega t)$. The measured and displayed magnitude would be 0.90 V (rms) ($1/\sqrt{2} \times 1.273$).

Degrees or Radians?

In this discussion, frequencies have been referred to as f (Hz) and ω ($2\pi f$ radians/sec). This is because people measure frequencies in cycles per second and math works best in radians. For purposes of measurement, frequencies as measured in a lock-in amplifier are in Hz. The equations used to explain the actual calculations are sometimes written using ω to simplify the expressions.

Phase is always reported in degrees. Once again, this is more by custom than by choice. Equations written as $\sin(\omega t + \theta)$ are written as if θ is in radians mostly for simplicity. Lock-in amplifiers always manipulate and measure phase in degrees.